

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum- 590014, Karnataka.



LAB RECORD

on

Big Data Analytics (23CS6PCBDA)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

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B.M.S. College of Engineering

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Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Big Data Analytics” carried out by **Yatharth Singh (1WA23CS051)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a Big Data Analytics (23CS6PCBDA) work prescribed for the said degree.

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INDEX

Sl. No.	Date	Experiment Title	Page No.
1	30.01.26	MongoDB- CRUD Operations Demonstration (Practice and Self Study)	5
2	06.02.26	Perform the following DB operations using Cassandra. • Create a keyspace by name Employee • Create a column family by name ◦ Employee-Info with attributes ◦ Emp_Id Primary Key, Emp_Name, Designation, ◦ Date_of_Joining, Salary, Dept_Name • Insert the values into the table in batch • Update Employee name and Department of EmpId 121 • Sort the details of Employee records based on salary • Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee. • Update the altered table to add project names. • Create a TTL of 15 seconds to display the values of Employees.	8
3	13.02.26	Perform the following DB operations using Cassandra. • Create a keyspace by name Library • Create a column family by name Library-Info with attributes ◦ Stud_Id Primary Key, ◦ Counter_value of type Counter, ◦ Stud_Name, Book-Name, Book-Id, ◦ Date_of_issue • Insert the values into the table in batch • Display the details of the table created and increase the value of the counter • Write a query to show that a student with id 112 has taken a book “BDA” 2 times. • Export the created column to a csv file • g) Import a given csv dataset from local file system into Cassandra column family	10
4	27.02.26	Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)	12
5	06.03.26	Implement Wordcount program on Hadoop framework	13

6	13.03.26	From the following link extract the weather data https://github.com/tomwhite/hadoop-book/tree/master/input/ncdc/all • Create a MapReduce program to find average temperature for each year from NCDC data set. • b) find the mean max temperature for every month	17
---	----------	---	----

7	10.04.26	For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.	22
8	10.04.26	Write a Scala program to print numbers from 1 to 100 using for loop.	29
9	10.04.26	Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.	30
10	2.04.26	Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).	31

Github Link: <https://github.com/yatharth-singh148/BDALab/tree/main>

Course Outcomes (COs):

CO1	Apply the concept of NoSQL, Hadoop or Spark for a given task
CO2	Analyse big data analytics mechanisms that can be applied to obtain solution for a given problem.
CO3	Design and implement solutions using data analytics mechanisms for a given problem.

LABORATORY PROGRAM – 1

MongoDB- CRUD Operations Demonstration (Practice and Self Study)

COMMAND WITH OUTPUT - USING ATLAS

```
test> use BDA_LAB
switched to db BDA_LAB
BDA_LAB> db.createCollection("Employee")
{ ok: 1 }
BDA_LAB> db.Employee.insertOne({
...   Emp_Id: 101,
...   Name: "Manoj",
...   Dept: "CSE",
...   Salary: 50000
... })
{
  acknowledged: true,
  insertedId: ObjectId('69f5bb2311d23b4927149f48')
}
BDA_LAB> db.Employee.insertMany([
...   {Emp_Id: 102, Name: "Rahul", Dept: "ISE", Salary: 40000},
...   {Emp_Id: 103, Name: "Anjali", Dept: "ECE", Salary: 45000},
...   {Emp_Id: 104, Name: "Kiran", Dept: "CSE", Salary: 55000}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f5bb2a11d23b4927149f49'),
    '1': ObjectId('69f5bb2a11d23b4927149f4a'),
    '2': ObjectId('69f5bb2a11d23b4927149f4b')
  }
}
BDA_LAB> db.Employee.find()
[
  {
    _id: ObjectId('69f5bb2311d23b4927149f48'),
    Emp_Id: 101,
    Name: 'Manoj',
```

```
}  
}  
BDA_LAB> db.Employee.find()  
[  
  {  
    _id: ObjectId('69f5bb2311d23b4927149f48'),  
    Emp_Id: 101,  
    Name: 'Manoj',  
    Dept: 'CSE',  
    Salary: 50000  
  },  
  {  
    _id: ObjectId('69f5bb2a11d23b4927149f49'),  
    Emp_Id: 102,  
    Name: 'Rahul',  
    Dept: 'ISE',  
    Salary: 40000  
  },  
  {  
    _id: ObjectId('69f5bb2a11d23b4927149f4a'),  
    Emp_Id: 103,  
    Name: 'Anjali',  
    Dept: 'ECE',  
    Salary: 45000  
  },  
  {  
    _id: ObjectId('69f5bb2a11d23b4927149f4b'),  
    Emp_Id: 104,  
    Name: 'Kiran',  
    Dept: 'CSE',  
    Salary: 55000  
  }  
]  
BDA_LAB> 
```

```
BDA_LAB> db.Employee.updateOne(
...   {Emp_Id: 101},
...   {$set: {Salary: 60000}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
BDA_LAB> db.Employee.updateMany(
...   {Dept: "CSE"},
...   {$set: {Dept: "AI"}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
BDA_LAB> █
```

LABORATORY PROGRAM – 2

Perform the following DB operations using Cassandra

Questions:

- a) Create a keyspace by name Employee
- b) Create a column family by name
 - Employee-Info with attributes
 - Emp_Id Primary Key, Emp_Name,
 - Designation, Date_of_Joining,
 - Salary, Dept_Name
- c) Insert the values into the table in batch
- d) Update Employee name and Department of Emp-Id 121
- e) Sort the details of Employee records based on salary
- f) Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
- g) Update the altered table to add project names.
- h) Create a TTL of 15 seconds to display the values of Employees.

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Employee
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};

cqlsh>
cqlsh> USE Employee;
cqlsh:employee> CREATE TABLE Employee_Info (
...     Emp_Id INT PRIMARY KEY,
...     Emp_Name TEXT,
...     Designation TEXT,
...     Date_of_Joining DATE,
...     Salary INT,
...     Dept_Name TEXT
... );
cqlsh:employee> BEGIN BATCH
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
... VALUES (121, 'Manoj', 'Engineer', '2022-01-10', 50000, 'CSE');
...
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
... VALUES (122, 'Rahul', 'Manager', '2021-05-15', 70000, 'ISE');
...
... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
... VALUES (123, 'Anjali', 'Analyst', '2023-03-20', 45000, 'ECE');
... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info
... SET Emp_Name = 'Manoj Kumar', Dept_Name = 'AI'
... WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info ORDER BY Salary DESC ALLOW FILTERING;
```

```
cqlsh:employee> ALTER TABLE Employee_Info
... ADD Projects SET<TEXT>;
cqlsh:employee> UPDATE Employee_Info
... SET Projects = {'Hadoop', 'Spark'}
... WHERE Emp_Id = 121;
cqlsh:employee>
cqlsh:employee> UPDATE Employee_Info
... SET Projects = {'ML', 'Data Mining'}
... WHERE Emp_Id = 122;
cqlsh:employee>
```



```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
... VALUES (130, 'TempUser', 'Intern', '2024-01-01', 20000, 'CSE')
... USING TTL 15;
cqlsh:employee> SELECT * FROM Employee_Info;
```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
123	2023-03-20	ECE	Analyst	Anjali	null	45000
130	2024-01-01	CSE	Intern	TempUser	null	20000
122	2021-05-15	ISE	Manager	Rahul	{ 'Data Mining', 'ML' }	70000
121	2022-01-10	AI	Engineer	Manoj Kumar	{ 'Hadoop', 'Spark' }	50000

(4 rows)

```
cqlsh:employee> SELECT * FROM Employee_Info;
```

emp_id	date_of_joining	dept_name	designation	emp_name	projects	salary
123	2023-03-20	ECE	Analyst	Anjali	null	45000
130	2024-01-01	CSE	Intern	TempUser	null	20000
122	2021-05-15	ISE	Manager	Rahul	{ 'Data Mining', 'ML' }	70000
121	2022-01-10	AI	Engineer	Manoj Kumar	{ 'Hadoop', 'Spark' }	50000

(4 rows)

LABORATORY PROGRAM – 3

Perform the following DB operations using Cassandra

Questions:

- a) Create a keyspace by name Library
- b) Create a column family by name Library-Info with attributes
 - Stud_Id Primary Key,
 - Counter_value of type Counter,
 - Stud_Name, Book-Name, Book-Id,
 - Date_of_issue
- c) Insert the values into the table in batch
- d) Display the details of the table created and increase the value of the counter
- e) Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
- f) Export the created column to a csv file
- g) Import a given csv dataset from local file system into Cassandra column family

COMMAND WITH OUTPUT

```
cqlsh> CREATE KEYSPACE Library
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
...     Stud_Id INT PRIMARY KEY,
...     Stud_Name TEXT,
...     Book_Name TEXT,
...     Book_Id INT,
...     Date_of_issue DATE
... );
cqlsh:library> CREATE TABLE Book_Counter (
...     Stud_Id INT,
...     Book_Name TEXT,
...     Counter_Value COUNTER,
...     PRIMARY KEY (Stud_Id, Book_Name)
... );
cqlsh:library> BEGIN BATCH
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
... VALUES (111, 'Manoj', 'BDA', 201, '2024-04-01');
...
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
... VALUES (112, 'Rahul', 'BDA', 202, '2024-04-02');
...
... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
... VALUES (113, 'Anjali', 'DBMS', 203, '2024-04-03');
... APPLY BATCH;
cqlsh:library> UPDATE Book_Counter
... SET Counter_Value = Counter_Value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Counter
... SET Counter_Value = Counter_Value + 1
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
```

```
cqlsh:library> SELECT * FROM Library_Info;
```

stud_id	book_id	book_name	date_of_issue	stud_name
111	201	BDA	2024-04-01	Manoj
113	203	DBMS	2024-04-03	Anjali
112	202	BDA	2024-04-02	Rahul

(3 rows)

```
cqlsh:library> SELECT * FROM Book_Counter;
```

stud_id	book_name	counter_value
112	BDA	2

(1 rows)

```
cqlsh:library> SELECT * FROM Book_Counter  
... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
```

stud_id	book_name	counter_value
112	BDA	2

(1 rows)

```
cqlsh:library> cqlsh -e "COPY Library.Library_Info TO 'library.csv' WITH HEADER = TRUE;"
```

```
...  
cqlsh:library> █
```

LABORATORY PROGRAM – 4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

COMMAND WITH OUTPUT

```
yatharth@yatharth-VirtualBox:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [yatharth-VirtualBox]
yatharth@yatharth-VirtualBox:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
yatharth@yatharth-VirtualBox:~$ cd Downloads
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -mkdir /input_file
yatharth@yatharth-VirtualBox:~/Downloads$ echo "Hello world!" > input.txt
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -put input.txt /input_file
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -ls /input_file
Found 1 items
-rw-r--r--  1 yatharth supergroup      13 2026-05-05 00:29 /input_file/input.txt
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -cat /input_file/input.txt
Hello world!
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -rm -r /input_file/input.txt
Deleted /input_file/input.txt
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -ls /input_file
yatharth@yatharth-VirtualBox:~/Downloads$ echo "Hello\nWorld" > input1.txt
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -cat /input1.txt
cat: `/input1.txt': No such file or directory
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -put input1.txt /input_file
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -cat /input_file/input1.txt
Hello\nWorld
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -rm -r /input_file
Deleted /input_file
yatharth@yatharth-VirtualBox:~/Downloads$
```

LABORATORY PROGRAM – 5

Implement Wordcount program on Hadoop framework

```
import java.io.Exception
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {

        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");

        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));

        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);

        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);

        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
    }

    // Main Method
```

```

    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}

```

Mapper Code:

```

// Importing libraries
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase
    implements Mapper<LongWritable, Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value,
        OutputCollector<Text, IntWritable> output,
        Reporter reporter) throws IOException {

        String line = value.toString();

        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
            if (word.length() > 0) {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}

```

Reducer Code

```

// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

```

```

public class WCReducer extends MapReduceBase
    implements Reducer<Text, IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> values,
        OutputCollector<Text, IntWritable> output,
        Reporter reporter) throws IOException {

        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
            count += values.next().get();
        }

        output.collect(key, new IntWritable(count));
    }
}

```

```

yatharth@yatharth-VirtualBox:~/Downloads
4728 ResourceManager
4819 NodeManager
4483 SecondaryNameNode
5799 Jps
4311 DataNode
4207 NameNode
yatharth@yatharth-VirtualBox:~/Downloads$ cd Downloads
yatharth@yatharth-VirtualBox:~/Downloads$ echo "hello hello this is hadoop in bda. bda = big data analysis." > input.txt
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -mkdir /inputf
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -put input.txt /inputf
yatharth@yatharth-VirtualBox:~/Downloads$ export HADOOP_CLASSPATH=$(hadoop classpath)
yatharth@yatharth-VirtualBox:~/Downloads$ javac --release 11 -classpath $(HADOOP_CLASSPATH) WordCount.java
HADOOP_CLASSPATH: command not found
error: no source files
yatharth@yatharth-VirtualBox:~/Downloads$ javac --release 11 -classpath $(HADOOP_CLASSPATH) WordCount.java
yatharth@yatharth-VirtualBox:~/Downloads$ jar -cvf wordcount.jar *.class
added manifest
adding: WordCount$IntSumReducer.class(in = 1755) (out= 750)(deflated 57%)
adding: WordCount$TokenizerMapper.class(in = 1873) (out= 837)(deflated 55%)
adding: WordCount.class(in = 1511) (out= 833)(deflated 44%)
yatharth@yatharth-VirtualBox:~/Downloads$ hadoop jar wordcount.jar WordCount /inputf /outputf
2026-05-05 00:41:05.096 INFO client.DefaultHadoopFalloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-05 00:41:06.216 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-05-05 00:41:06.257 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yatharth/.staging/job_1777921102769_0001
2026-05-05 00:41:06.726 INFO input.FileInputFormat: Total input files to process : 1
2026-05-05 00:41:07.272 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-05 00:41:07.700 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777921102769_0001
2026-05-05 00:41:07.700 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-05 00:41:08.026 INFO conf.Configuration: resource-types.xml not found
2026-05-05 00:41:08.031 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'
2026-05-05 00:41:10.366 INFO impl.YarnClientImpl: Submitted application application_1777921102769_0001
2026-05-05 00:41:10.557 INFO mapreduce.Job: The url to track the job: http://yatharth-VirtualBox:8088/proxy/application_1777921102769_0001/
2026-05-05 00:41:10.566 INFO mapreduce.Job: Running job: job_1777921102769_0001
2026-05-05 00:41:28.454 INFO mapreduce.Job: Job job_1777921102769_0001 running in uber mode : false
2026-05-05 00:41:28.466 INFO mapreduce.Job: map 0% reduce 0%
2026-05-05 00:41:42.061 INFO mapreduce.Job: map 100% reduce 0%
2026-05-05 00:41:48.139 INFO mapreduce.Job: map 100% reduce 100%
2026-05-05 00:41:49.164 INFO mapreduce.Job: Job job_1777921102769_0001 completed successfully
2026-05-05 00:41:49.317 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=126
FILE: Number of bytes written=551777
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=163
HDFS: Number of bytes written=76

```

```

Reduce input groups=11
Reduce shuffle bytes=126
Reduce input records=11
Reduce output records=11
Spilled Records=22
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=116
CPU time spent (ms)=3530
Physical memory (bytes) snapshot=515768320
Virtual memory (bytes) snapshot=5467938816
Total committed heap usage (bytes)=349175808
Peak Map Physical memory (bytes)=315756544
Peak Map Virtual memory (bytes)=2729398272
Peak Reduce Physical memory (bytes)=200011776
Peak Reduce Virtual memory (bytes)=2738540544
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=60
File Output Format Counters
  Bytes Written=76
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -ls /outputf
Found 2 items
-rw-r--r--  1 yatharth supergroup          0 2026-05-05 00:41 /outputf/_SUCCESS
-rw-r--r--  1 yatharth supergroup        76 2026-05-05 00:41 /outputf/part-r-00000
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -cat /outputf/part-r-00000
=      1
analysis.      1
bda      1
bda.      1
big      1
data      1
hadoop      1
hello      2
in      1
is      1
this      1
yatharth@yatharth-VirtualBox:~/Downloads$ █

```


LABORATORY PROGRAM – 6

Implement Weather program on Hadoop framework

Questions:

From the following link extract the weather data

<https://github.com/tomwhite/hadoopbook/tree/master/input/ncdc/all>

- a) Create a MapReduce program to find average temperature for each year from NCDC data set.
- b) find the mean max temperature for every month.

CODE, COMMAND WITH OUTPUT – A Driver Code

```
package temp;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Please enter both input and output parameters.");
            System.exit(-1);
        }

        // Creating a configuration and job instance
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Average Calculation");
        job.setJarByClass(AverageDriver.class);

        // Input and output paths
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));

        // Setting mapper and reducer classes
        job.setMapperClass(AverageMapper.class);
        job.setReducerClass(AverageReducer.class);
```

```

    // Output key and value types
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    // Submitting the job and waiting for it to complete
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

MAPPER CODE

```

package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> {

    public static final int MISSING = 9999;

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Extract year from fixed position
        String year = line.substring(15, 19);
        int temperature;

        // Determine if there's a '+' sign
        if (line.charAt(87) == '+') {

```

```

        temperature = Integer.parseInt(line.substring(88, 92));
    } else {
        temperature = Integer.parseInt(line.substring(87, 92));
    }

    // Quality check character
    String quality = line.substring(92, 93);

    // Only emit if data is valid
    if (temperature != MISSING && quality.matches("[01459]")) {
        context.write(new Text(year), new IntWritable(temperature));
    }
}

```

REDUCER CODE

```

package temp;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sumTemp = 0;
        int count = 0;
    }
}

```

```

for (IntWritable value : values) {
    sumTemp += value.get();
    count++;
}

if (count > 0) {
    int average = sumTemp / count;
    context.write(key, new IntWritable(average));
}
}
}

```

```

yatharth@yatharth-VirtualBox: ~/Downloads$ nano weather.txt
yatharth@yatharth-VirtualBox: ~/Downloads$ nano weather.txt
yatharth@yatharth-VirtualBox: ~/Downloads$ hdfs dfs -mkdir /inputf
yatharth@yatharth-VirtualBox: ~/Downloads$ hdfs dfs -put weather.txt /inputf
yatharth@yatharth-VirtualBox: ~/Downloads$ export HADOOP_CLASSPATH=$(hadoop classpath)
yatharth@yatharth-VirtualBox: ~/Downloads$ javac --release 11 -classpath ${HADOOP_CLASSPATH} MeanMaxMonth.java
yatharth@yatharth-VirtualBox: ~/Downloads$ javac --release 11 -classpath ${HADOOP_CLASSPATH} MeanMaxMonth.java
yatharth@yatharth-VirtualBox: ~/Downloads$ jar -cvf meanmaxmonth.jar *.class
added manifest
adding: MeanMaxMonth$MeanReducer.class(in = 1810) (out= 778)(deflated 57%)
adding: MeanMaxMonth$MonthMapper.class(in = 2024) (out= 899)(deflated 55%)
adding: MeanMaxMonth.class(in = 1659) (out= 922)(deflated 44%)
yatharth@yatharth-VirtualBox: ~/Downloads$ hadoop jar meanmaxmonth.jar MeanMaxMonth /inputf /outputf
2026-05-05 01:09:16,335 INFO client.DefaultNoHARMFaileoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-05 01:09:17,266 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interf
2026-05-05 01:09:17,346 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yatharth/.stagi
2026-05-05 01:09:17,825 INFO input.FileInputFormat: Total input files to process : 1
2026-05-05 01:09:18,452 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-05 01:09:18,799 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777921102769_0002
2026-05-05 01:09:18,800 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-05 01:09:19,036 INFO conf.Configuration: resource-types.xml not found
2026-05-05 01:09:19,037 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-05 01:09:19,955 INFO impl.YarnClientImpl: Submitted application application_1777921102769_0002
2026-05-05 01:09:20,038 INFO mapreduce.Job: The url to track the job: http://yatharth-VirtualBox:8088/proxy/application_1777921102769_0
2026-05-05 01:09:20,039 INFO mapreduce.Job: Running job: job_1777921102769_0002
2026-05-05 01:09:31,711 INFO mapreduce.Job: Job job_1777921102769_0002 running in uber mode : false
2026-05-05 01:09:31,712 INFO mapreduce.Job: map 0% reduce 0%
2026-05-05 01:09:37,911 INFO mapreduce.Job: map 100% reduce 0%
2026-05-05 01:09:43,988 INFO mapreduce.Job: map 100% reduce 100%
2026-05-05 01:09:44,021 INFO mapreduce.Job: Job job_1777921102769_0002 completed successfully
2026-05-05 01:09:44,147 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=71
  FILE: Number of bytes written=551351
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=145
  HDFS: Number of bytes written=24
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

```

```
Reduce input records=5
Reduce output records=3
Spilled Records=10
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=59
CPU time spent (ms)=1610
Physical memory (bytes) snapshot=508702720
Virtual memory (bytes) snapshot=5468119040
Total committed heap usage (bytes)=466616320
Peak Map Physical memory (bytes)=277741568
Peak Map Virtual memory (bytes)=2731675648
Peak Reduce Physical memory (bytes)=230961152
Peak Reduce Virtual memory (bytes)=2736443392
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=40
File Output Format Counters
  Bytes Written=24
```

```
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -ls /outputf
```

```
Found 2 items
```

```
-rw-r--r--  1 yatharth supergroup      0 2026-05-05 01:09 /outputf/_SUCCESS
-rw-r--r--  1 yatharth supergroup    24 2026-05-05 01:09 /outputf/part-r-00000
```

```
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -cat /outputf/part-r-00000
```

```
01      24.2
```

```
02      19.0
```

```
03      30.0
```

```
yatharth@yatharth-VirtualBox:~/Downloads$ █
```

LABORATORY PROGRAM – 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words.

```
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopNDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 3) {
            System.err.println("Usage: TopNDriver <in> <temp-out> <final-out>");
            System.exit(2);
        }

        Configuration conf = new Configuration();

        // === Job 1: Word Count ===
        Job wcJob = Job.getInstance(conf, "word count");
        wcJob.setJarByClass(TopNDriver.class);
        wcJob.setMapperClass(WordCountMapper.class);
        wcJob.setCombinerClass(WordCountReducer.class);
        wcJob.setReducerClass(WordCountReducer.class);
        wcJob.setOutputKeyClass(Text.class);
        wcJob.setOutputValueClass(IntWritable.class);

        FileInputFormat.addInputPath(wcJob, new Path(args[0]));
        Path tempDir = new Path(args[1]);
        FileOutputFormat.setOutputPath(wcJob, tempDir);

        if (!wcJob.waitForCompletion(true)) {
            System.exit(1);
        }

        // === Job 2: Top N ===
```

```

Job topJob = Job.getInstance(conf, "top 10 words");
topJob.setJarByClass(TopNDriver.class);
topJob.setMapperClass(TopNMapper.class);
topJob.setReducerClass(TopNReducer.class);
topJob.setMapOutputKeyClass(IntWritable.class);
topJob.setMapOutputValueClass(Text.class);
topJob.setOutputKeyClass(Text.class);
topJob.setOutputValueClass(IntWritable.class);

FileInputFormat.addInputPath(topJob, tempDir);
FileOutputFormat.setOutputPath(topJob, new Path(args[2]));

System.exit(topJob.waitForCompletion(true) ? 0 : 1);
}
}

```

Mapper Code (WordCountMapper.java)

```

package samples.topn;

import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class WordCountMapper extends Mapper<Object, Text, Text, IntWritable> {

    private static final IntWritable ONE = new IntWritable(1);
    private Text word = new Text();

    // characters to normalize into spaces
    private String tokens = "[_!$%&^*~\[\]\|\\.\,:;'\"]";

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // clean & tokenize
        String clean = value.toString()
            .toLowerCase()
            .replaceAll(tokens, " ");

        StringTokenizer itr = new StringTokenizer(clean);
    }
}

```

```

        while (itr.hasMoreTokens()) {
            word.set(itr.nextToken().trim());
            context.write(word, ONE);
        }
    }
}

```

Mapper Code (TopNMapper.java)

```

package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TopNMapper extends Mapper<Object, Text, IntWritable, Text> {

    private IntWritable count = new IntWritable();
    private Text word = new Text();

    @Override
    protected void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        // input line: word \t count
        String[] parts = value.toString().split("\\t");

        if (parts.length == 2) {
            word.set(parts[0]);
            count.set(Integer.parseInt(parts[1]));

            // emit count → word, so Hadoop sorts by count
            context.write(count, word);
        }
    }
}

```

Reducer Code (WordCountReducer.java)

```

package samples.topn;

import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

```



```

public class WordCountReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    protected void reduce(Text key, Iterable<IntWritable> values, Context context)
        throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable val : values) {
            sum += val.get();
        }

        context.write(key, new IntWritable(sum));
    }
}

```

Reducer Code (TopNReducer.java)

```

package samples.topn;

import java.io.IOException;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Map;
import java.util.TreeMap;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopNReducer extends Reducer<IntWritable, Text, Text, IntWritable> {

    // TreeMap with descending order of keys (counts)
    private TreeMap<Integer, List<String>> countMap =
        new TreeMap<>(Collections.reverseOrder());

    @Override
    protected void reduce(IntWritable key, Iterable<Text> values, Context context)
        throws IOException, InterruptedException {

        int cnt = key.get();

        List<String> words = countMap.getOrDefault(cnt, new ArrayList<>());
    }
}

```

```

    for (Text w : values) {
        words.add(w.toString());
    }

    countMap.put(cnt, words);
}

@Override
protected void cleanup(Context context)
    throws IOException, InterruptedException {

    // collect top 10 word→count pairs
    List<WordCount> topList = new ArrayList<>();
    int seen = 0;

    for (Map.Entry<Integer, List<String>> entry : countMap.entrySet()) {
        int cnt = entry.getKey();

        for (String w : entry.getValue()) {
            topList.add(new WordCount(w, cnt));
            seen++;

            if (seen == 10) break;
        }

        if (seen == 10) break;
    }

    // emit results
    for (WordCount wc : topList) {
        context.write(new Text(wc.getWord()), new IntWritable(wc.getCount()));
    }
}

// sort these 10 entries alphabetically by word
Collections.sort(topList, (a, b) -> a.word.compareTo(b.word));

// emit final top 10 in alphabetical order
for (WordCount wc : topList) {
    context.write(new Text(wc.word), new IntWritable(wc.count));
}

}

// helper class

```

```

private static class WordCount {
    String word;
    int count;

    WordCount(String w, int c) {
        word = w;
        count = c;
    }
}
}

```

```

yatharth@yatharth-VirtualBox:~/Downloads$ nano input.txt
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -mkdir /inputf
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -put input.txt /inputf
yatharth@yatharth-VirtualBox:~/Downloads$ javac --release 11 -classpath ${HADOOP_CLASSPATH} Top10Words.java
Top10Words.java:32: error: <identifier> expected
Alphabetically
^
1 error
yatharth@yatharth-VirtualBox:~/Downloads$ javac --release 11 -classpath ${HADOOP_CLASSPATH} Top10Words.java
yatharth@yatharth-VirtualBox:~/Downloads$ jar -cvf top10words.jar *.class
added manifest
adding: Top10Words$TokenizerMapper.class(in = 1880) (out= 834)(deflated 55%)
adding: Top10Words$Top10Reducer.class(in = 3506) (out= 1428)(deflated 59%)
adding: Top10Words.class(in = 1521) (out= 844)(deflated 44%)
yatharth@yatharth-VirtualBox:~/Downloads$ hadoop jar top10words.jar Top10Words /inputf /outputf
2026-05-05 01:26:39,747 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-05-05 01:26:40,113 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute you
2026-05-05 01:26:40,150 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/yatharth/.staging/job_177792110276
2026-05-05 01:26:40,491 INFO input.FileInputFormat: Total input files to process : 1
2026-05-05 01:26:40,621 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-05 01:26:40,822 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1777921102769_0003
2026-05-05 01:26:40,822 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-05 01:26:40,993 INFO conf.Configuration: resource-types.xml not found
2026-05-05 01:26:40,994 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-05-05 01:26:41,110 INFO impl.YarnClientImpl: Submitted application application_1777921102769_0003
2026-05-05 01:26:41,189 INFO mapreduce.Job: The url to track the job: http://yatharth-VirtualBox:8088/proxy/application_1777921102769_0003/
2026-05-05 01:26:41,190 INFO mapreduce.Job: Running job: job_1777921102769_0003
2026-05-05 01:26:49,456 INFO mapreduce.Job: Job job_1777921102769_0003 running in uber mode : false
2026-05-05 01:26:49,458 INFO mapreduce.Job: map 0% reduce 0%
2026-05-05 01:26:54,562 INFO mapreduce.Job: map 100% reduce 0%
2026-05-05 01:27:01,679 INFO mapreduce.Job: map 100% reduce 100%
2026-05-05 01:27:01,697 INFO mapreduce.Job: Job job_1777921102769_0003 completed successfully
2026-05-05 01:27:01,850 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=325
  FILE: Number of bytes written=551835
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=260
  HDFS: Number of bytes written=41
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2

```

```
Peak Map Physical memory (bytes)=260943872
Peak Map Virtual memory (bytes)=2732863488
Peak Reduce Physical memory (bytes)=200785920
Peak Reduce Virtual memory (bytes)=2743091200
```

Shuffle Errors

```
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
```

File Input Format Counters

```
Bytes Read=157
```

File Output Format Counters

```
Bytes Written=41
```

```
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -ls /outputf
```

Found 2 items

```
-rw-r--r--  1 yatharth supergroup      0 2026-05-05 01:26 /outputf/_SUCCESS
-rw-r--r--  1 yatharth supergroup    41 2026-05-05 01:26 /outputf/part-r-00000
```

```
yatharth@yatharth-VirtualBox:~/Downloads$ hdfs dfs -cat /outputf/part-r-00000
```

hadoop : 4

hello : 3

is : 2

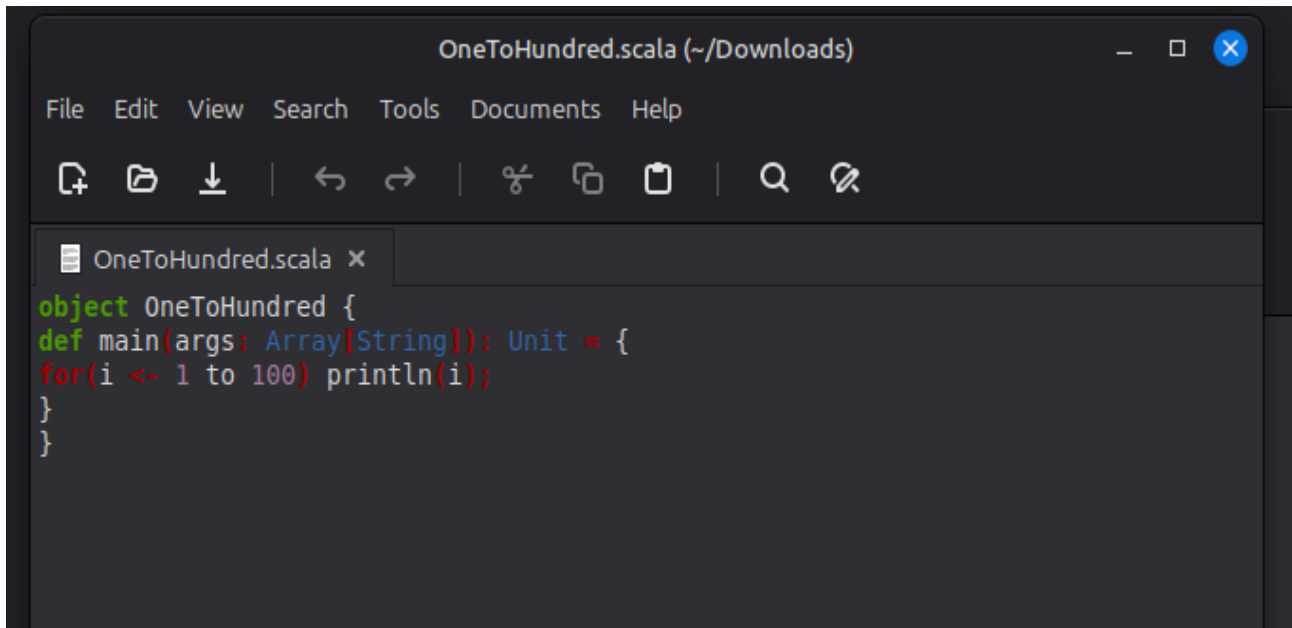
yatharth : 1

```
yatharth@yatharth-VirtualBox:~/Downloads$
```

LABORATORY PROGRAM – 8

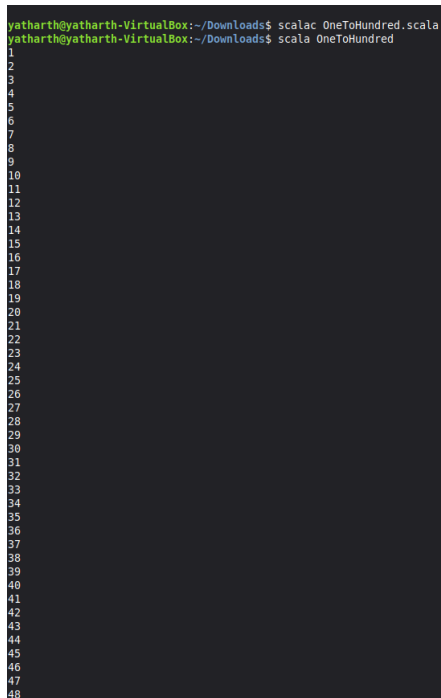
Write a Scala program to print numbers from 1 to 100 using for loop.

CODE, COMMAND WITH OUTPUT



The screenshot shows a Scala IDE window titled "OneToHundred.scala (~/.Downloads)". The menu bar includes File, Edit, View, Search, Tools, Documents, and Help. The toolbar contains icons for file operations and search. The code editor displays the following Scala code:

```
object OneToHundred {  
  def main(args: Array[String]): Unit = {  
    for(i <- 1 to 100) println(i);  
  }  
}
```



The screenshot shows a terminal window with the following commands and output:

```
yatharth@yatharth-VirtualBox:~/Downloads$ scalac OneToHundred.scala  
yatharth@yatharth-VirtualBox:~/Downloads$ scala OneToHundred  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
21  
22  
23  
24  
25  
26  
27  
28  
29  
30  
31  
32  
33  
34  
35  
36  
37  
38  
39  
40  
41  
42  
43  
44  
45  
46  
47  
48
```

```
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72
73
74
75
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
```

```
yatharth@yatharth-VirtualBox:~/Downloads$
```

LABORATORY PROGRAM – 9

Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

CODE, COMMAND WITH OUTPUT

```
>>> rdd = spark.sparkContext.textFile("input.txt")
>>> words = rdd.flatMap(lambda line: line.split(" "))
>>> pairs = words.map(lambda word: (word, 1))
>>> counts = pairs.reduceByKey(lambda a, b: a + b)
>>> result = counts.filter(lambda x: x[1] > 4)
>>> result.collect()
[('hello', 5), ('spark', 6)]
```

LABORATORY PROGRAM – 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

CODE, COMMAND WITH OUTPUT

```
# Install NLTK and download required data (run once)
!pip install nltk

import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

from pyspark.sql import SparkSession
from pyspark.sql.functions import col, lower, regexp_replace, split, explode, udf
from pyspark.sql.types import ArrayType, StringType
from pyspark.ml.feature import StopWordsRemover
from nltk.stem import WordNetLemmatizer

# Initialize SparkSession
spark = SparkSession.builder.appName("TextProcessing").getOrCreate()

# Define your input lines
lines = [
    "Hello, I hate you.",
    "I hate that I love you.",
    "Don't want to, but I can't put",
    "nobody else above you."
]

# Create DataFrame from lines
df = spark.createDataFrame(lines, "string").toDF("value")

# Step 1: Lowercase and remove punctuation
df_clean = df.select(
    regexp_replace(lower(col("value")), "[^a-zA-Z\\s]", "").alias("cleaned")
)
```



```

# Step 2: Tokenize the cleaned text
df_tokens = df_clean.select(
    split(col("cleaned"), "\\s+").alias("tokens")
)

# Step 3: Remove stop words
remover = StopWordsRemover(inputCol="tokens", outputCol="filtered")
df_filtered = remover.transform(df_tokens)

# Step 4: Lemmatization using NLTK WordNetLemmatizer with UDF
lemmatizer = WordNetLemmatizer()

def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

lemmatize_udf = udf(lemmatize_words, ArrayType(StringType()))

df_lemmatized = df_filtered.withColumn(
    "lemmatized",
    lemmatize_udf(col("filtered"))
)

# Step 5: Explode the lemmatized words and show results
df_lemmatized.select(
    explode(col("lemmatized")).alias("word")
).show(truncate=False)

```

